# Evaluation of Temperature Rise and Reliability for DC-Link Capacitors in Traction Inverters by Synergy of Lumped-Parameter and FEA Thermal Models

Kaining Kuang, *Graduate Student Member, IEEE*, Jianjun Kuang, Zhengyan Zhou, and Jiachen Tian

***Abstract*— Film capacitors are used in traction inverters to stabilize the voltage on the battery side. However, excessive hotspot temperatures can lead to premature capacitor failure, significantly reducing the reliability of the entire traction system. Monitoring the temperature rise profile of capacitors is vital for assessing their lifetime. Existing finite element analysis (FEA) and lumped thermal network (LTN) models for capacitors have certain limitations: FEA requires substantial computational resources, while traditional LTN models fail to account for the influence of heat capacity on thermal dynamics, which limits their effectiveness in temperature monitoring. To address this issue, this article develops a second-order LTN, whose parameters are obtained through FEA, to incorporate the effect of heat capacity on the thermal characteristics of capacitors. Subsequently, the proposed LTN is combined with the ripple current profile of an electric vehicle (EV) to efficiently calculate the capacitor's hotspot temperature. Based on this, the reliability of capacitors is evaluated using cumulative fatigue damage theory. The findings provide insights for planning the scheduled maintenance of capacitors used in traction inverters.**

***Index Terms*— Electric vehicle (EV), film capacitor, finite element method, lumped thermal network (LTN), reliability evaluation.**

## I. Introduction

IN THE traction inverter of vehicles, the limited switching frequency of power modules often results in ripple voltages and currents. These ripples can cause repeated polarity reversals between the electrolyte and electrodes in the battery, ultimately leading to degradation of battery performance. To mitigate these effects, film capacitors with low equivalent series resistance (ESR) and good frequency characteristics are widely used to absorb ripple currents and stabilize voltage [1], [2]. However, as shown in Fig. 1(a), the maximum operating temperature of film capacitors is lower than that of other inverter components such as digital signal processor (DSP) modules, gate drivers and power modules [3], [4]. Fig. 1(b) further shows that capacitors have the highest failure rates among passive components in traction inverters, triggering one-fifth of all faults [5], [6]. Although capacitor failure is affected by factors including temperature, voltage, and humidity, temperature is widely recognized as the most critical contributor, according to the International Electrotechnical Commission (IEC) standard [7]. However, due to the complex packaging structure of film capacitors, it is virtually impossible to measure hotspot temperature directly in laboratory settings. As a result, many researchers use simulations rather than experiments to assess temperature rise. For instance, [8] developed a micro-scale finite element analysis (FEA) model of polymer film to extract equivalent material parameters, which were then used in a macroscale model to evaluate the hotspot temperature of the capacitor, limiting the error to within 5%. Kuang et al. [9] pointed out the nonuniform loss distribution within the capacitor core and proposed a layered model to capture this distribution and improve simulation accuracy. Based on such discrete capacitor thermal models, [2] examined thermal interactions within capacitor banks and suggested reducing temperature rise by adjusting the capacitor arrangement layout. Likewise, [3] investigated how geometric parameters influence temperature rise within capacitor banks. Niu et al. [10] emphasized that heat from other converter components affects capacitor temperature, and built a multilevel FEA of the converter to analyze thermal coupling between power modules and capacitors while maintaining acceptable computational efficiency.

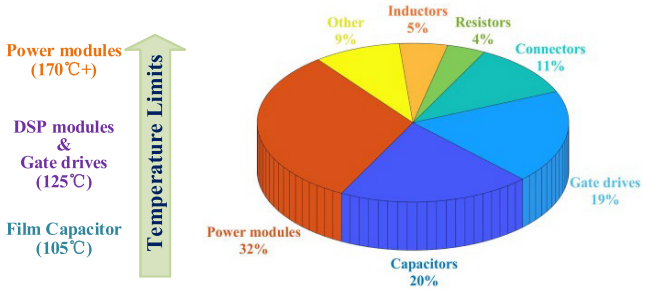



Fig. 1. Thermal limits and failure characteristics of components in traction inverters. (a) Upper temperature limits of components used in traction inverters. (b) Failure probability of each component in inverters.

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However, all the above methods heavily rely on FEA, which is computationally intensive. Even for steady-state analysis of a single capacitor under fixed power loss conditions, the simulation can take hundreds of seconds. In electric vehicles (EVs), where capacitor losses change rapidly and driving durations often span thousands of seconds, applying FEA for transient thermal analysis would require substantial computational resources. Therefore, these methods are not directly suitable for evaluating the temperature rise of capacitors in EVs or other time-dependent applications. To address this, some researchers have adopted lumped thermal networks (LTNs), which transform the heat conduction process into a set of thermal nodes to enable faster analysis: [11] calculated the thermal resistance of capacitors under various physical configurations and developed an LTN to assess capacitor thermal stress. Similarly, [12] proposed a comparable network to estimate capacitors' temperature and compare thermal stress under different control strategies. Yin et al. [13] further improved the LTN by extending it into a thermal resistance matrix, allowing for rapid evaluation of thermal coupling among capacitors. Building on these LTNs, several studies have examined long-term thermal stress in capacitors under task profiles. For instance, [14] highlighted that many LTNs overlook ambient temperature fluctuations and proposed a method to evaluate annual temperature variation using real environmental data. Yang et al. [15] and Zhou et al. [16] incorporated power loss variations from photovoltaic and wind power systems into their models to obtain more realistic temperature profiles for capacitors. However, since manufacturers typically only provide approximate thermal resistance values between the capacitor hotspot and the ambient without specifying thermal capacitance, most LTNs are simplified to first-order thermal resistance networks. This simplification leads to unrealistic temperature transitions that change instantly with current fluctuations. In reality, capacitors take thousands of seconds to reach thermal equilibrium. Such simplifications can result in overestimations of temperature rise, potentially leading to unnecessary capacitor replacement [14]. Thus, improving LTN models by considering thermal capacitance is essential.

Based on the thermal stress profile, some studies have applied damage accumulation models to evaluate the lifetime of capacitors. Huang et al. [17] and Wang et al. [18] assessed electro-thermal stress in drive system capacitors and estimated their operational lifetimes. Lenz et al. [19] conducted a similar investigation for photovoltaic systems, examining the influence of switching frequency on capacitor aging. Wang et al. [20] analyzed the impact of ESR variation on temperature rise and degradation and proposed a capacitor lifetime model that accounts for accelerated aging effects. These studies found significant variations in capacitor lifetime under different applications and control strategies, demonstrating the need to incorporate temperature rise profiles in reliability assessments. Yao et al. [21] compared the dc-link capacitor bank currents under different modulation strategies in a railway traction drive system, verified the optimization effectiveness through capacitor case-temperature monitoring experiments, and evaluated the capacitor lifetime. Yao et al. [22] conducted a similar study and further incorporated the annual variation of ambient temperature into the model to enable a more accurate assessment of capacitor lifetime. However, research explicitly focusing on the reliability of dc-link capacitors in EVs is still scarce, and the distinct differences in thermal dissipation design and loss characteristics between previous capacitor banks used in high-power applications and those in EVs hinder the direct applicability of these methods to EV traction inverters. Given the frequent and significant ripple current fluctuations in EV applications, reliability assessments based on real-world task profiles are essential for predictive maintenance, reducing unexpected system failures, and improving the overall reliability of electric drive systems [23].

Therefore, this article primarily addresses the issue that existing LTN models for capacitors neglect the impact of heat capacity, which affects the accuracy of temperature rise predictions and reliability assessments. The main contributions of this article are as follows.

1) The ripple current profile of dc-link capacitors used in EVs is obtained from a Simulink model of the motor system based on the ECE-15 urban driving cycle.
2) An FEA model of film capacitors is developed and experimentally validated. On this basis, a second-order *RC* thermal network of the capacitor is constructed to enable rapid tracking of variations in the hotspot temperature of capacitors.
3) The reliability of the capacitor is assessed based on Miner's rule and the IEC 61709 standard. The Monte Carlo method is used to account for the impact of manufacturing tolerances on capacitor reliability, and a two-parameter Weibull function is employed to fit the failure distribution and determine the B10 life of the capacitor, enabling predictive maintenance.

## II. Analysis of Film Capacitors for EV Application

This section presents the structure, parameters, model selection criteria, and failure mechanisms of capacitors. These analyses will form the basis for the simulations and reliability assessments discussed in Sections III and IV.

### A. Structure and Parameters of Film Capacitors

The core of film capacitors is made by winding two layers of metallized polypropylene film. Metal layers are vapor-deposited onto the polymer film surfaces, forming two nonpolarized terminals. The wound core and connecting pins are bonded using a zinc-spray coating and enclosed in a polyphenylene sulfide casing. The space between the core and casing is filled with epoxy resin for mechanical support and electrical insulation [8], [20]. Fig. 2 shows the structural and materials of the film capacitor, along with the disassembled capacitor core.

The specifications of the 110 $\mu$F Faratronics C3D capacitor used in this study are summarized in Table I [24]. Specifically, the nominal capacitance is 110 $\mu$F, with a permissible deviation of $\pm 10\%$ from the rated value. The rated voltage, defined as the maximum dc or peak pulse voltage that can be continuously applied, is 500 V. In terms of thermal characteristics, the thermal resistance between the capacitor hotspot and the ambient is specified as 10 °C/W, while the operational temperature range of $-40$ °C to 105 °C specifies the permissible combination of ambient temperature and internal self-heating. Furthermore, the rated voltage slew rate of 15 V/$\mu$s, when multiplied by the capacitance of 110 $\mu$F,

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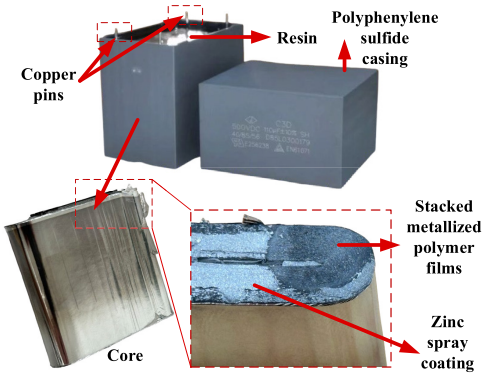



Fig. 2. Structure and materials of the 110 $\mu$F C3D film capacitor.

TABLE I
PARAMETERS OF THE C3D CAPACITOR

| Parameters | Values |
|---|---|
| Capacitance (μF) | 110 (±10%) |
| Rated voltage (V) | 500 |
| Expected lifetime (h) | 100 000h ($T_h$=70°C) |
| Size (mm) | 57 (width), 50 (height), 35 (thickness) |
| $R_{th}$ (°C/W) | 10 |
| ESR (Ω) | 0.013 (50 Hz) |
| Operating temperature (°C) | -40-105 |
| Voltage slew rate (V/μs) | 15 |
| Time constant (s) | ≥10000 |

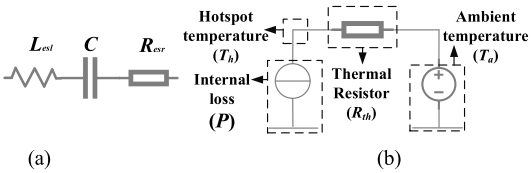



Fig. 3. Lumped model of capacitors. (a) Electrical model. (b) Thermal model.

yields an instantaneous peak current capability of 1650 A, reflecting the capacitor's capacity to handle transient current surges on the sub-millisecond timescale. In addition, the insulation resistance, expressed as the time constant $\tau$ (the product of insulation resistance and capacitance), is specified to be ≥10 000 s, which corresponds to a resistance of ≥90.91 MΩ.

Fig. 3 shows the equivalent circuit of the capacitor. In Fig. 3(a), $C$ represents the capacitance ($\mu$F), $R_{\text{esr}}$ is the ESR (mΩ), and $L_{esl}$ refers to the equivalent series inductance of the capacitor (nH). For dc-Link film capacitors, to reduce oscillations at switching frequencies, $L_{esl}$ is typically below 100 nH [5]. Therefore, the equivalent circuit of a dc-Link capacitor can be further simplified as a series combination of $C$ and $R_{\text{esr}}$. Since the external epoxy resin and the PPS housing do not contribute to power dissipation, the capacitor core is regarded as the primary heat source of the film capacitor. The power losses of the capacitor core are composed of electrode losses, dielectric losses, and leakage losses. Among these loss components, leakage loss refers to the power dissipation caused by the leakage current flowing through the insulation resistance under dc bias. Considering that the dc voltage limit of the employed capacitor is 500 V, while the insulation resistance is 90.91 MΩ, the corresponding leakage power is less than 0.003 W. Dielectric loss arises from the internal

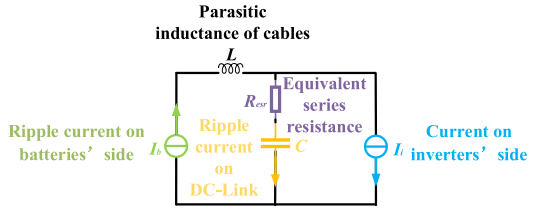



Fig. 4. Equivalent circuit of ripple current analysis for motor systems.

energy dissipation of the dielectric material under an alternating electric field, which is mainly associated with polarization processes of the molecular structure of the dielectric. Electrode loss, on the other hand, results from Joule heating due to the current flowing through the metallized electrodes. In practical applications, the total power loss is often estimated by multiplying the square of the current through the capacitor by its ESR in Fig. 3(a)

$$P = I^2 \times R_{\text{esr}}. \tag{1}$$

The generated heat is concentrated in the capacitor core and subsequently dissipates through the resin and casing into the surrounding environment. For temperature rise evaluation, researchers typically adopt the pure LTN model of capacitors, as depicted in Fig. 3(b). This LTN is analogous to an electrical circuit: the ambient temperature ($T_a$,°C) acts as a voltage source, while internal power losses ($P$, W) are modeled as a current source supplying power to the LTN. Thermal resistance ($R_{\text{th}}$,°C/W), similar to electrical resistance, is defined as the temperature difference between the hotspot and the ambient divided by the loss value. Since thermal resistance just varies slightly with ambient temperature, capacitor manufacturers generally only provide a fixed representative value for the thermal resistance between the capacitor hotspot and the ambient, which remains approximately valid across the normal operating temperature range. Based on this, the hotspot temperature ($T_h$,°C) can be calculated analogously to using Ohm's Law to determine voltage [10]

$$T_{\text{h}} = P \times R_{\text{th}} + T_{\text{a}}. \tag{2}$$

To effectively suppress ripple current and stabilize the dc-link voltage, capacitor selection must consider both current-handling capacity and voltage fluctuation mitigation. Regarding current absorption, the dc-link capacitor is expected to ensure that the ripple current reaching the battery side ($I_b$, $A$) remains below 10% of its rated current. Fig. 4 shows the simplified equivalent circuit used for ripple current analysis in motor systems, in which the power module and load are treated as a current source ($I_i$, $A$) [25].

In EVs, the battery and inverter are often placed at opposite ends of the vehicle, resulting in long connecting cables that introduce several mH of parasitic inductance ($L$). Based on this, $I_b$ can be quantitatively related to the capacitance value and the ESR of the capacitor

$$I_b(s) = I_i(s) \times \frac{R_{\text{esr}}Cs + 1}{LCs^2 + R_{\text{esr}}Cs + 1}. \tag{3}$$

Regarding voltage suppression, capacitors should limit voltage ripple to within 2% of the battery voltage to reduce potential risks. Under the common space vector pulsewidth modulation (SVPWM) strategy, the relationship between

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TABLE II
FAILURE MODES OF FILM CAPACITORS

| Failure mode | Failure principle | Related factors |
|---|---|---|
| Aging failure | Parameter drift in capacitance and ESR lead by self-healing effects and capacitor aging | voltage, ripple current, temperature, humidity |
| Open circuit fault | Disconnection between the core and terminals | ripple current, temperature, vibration |
| Short circuit fault | Breakdown of films under high voltage, core melting due to high current | voltage, ripple current, temperature |

capacitance and voltage ripple can be described by

$$\Delta u = \frac{C f_s M}{I_N 16} \times \sqrt{\left(6 - M\frac{96\sqrt{3}}{5\pi} + M^2\frac{108\pi - 81\sqrt{3}}{5\pi}\right)\cos\varphi^2 + M\frac{8\sqrt{3}}{5\pi}} \tag{4}$$

where $f_s$ is the switching frequency (Hz), $I_N$ is the inverter output current ($A$), $\Delta u$ is the voltage ripple ($V$), $M$ is the modulation index, and $\cos\varphi$ is the power factor of the load.

### B. Failure Analysis of Film Capacitors

As shown in Table II, the failure modes of film capacitors can be broadly categorized into catastrophic and aging failures, with aging failures being the predominant mode in most applications. Catastrophic failures include short circuits, typically caused by dielectric breakdown, and open circuits, which often result from overheating or solder joint detachment. Aging failures, on the other hand, occur due to the gradual degradation of materials under prolonged thermal and electrical stress. This degradation is mainly caused by polymer chain scission and crosslinking in the dielectric material, as well as oxidation and corrosion of the metallized electrode layers. These processes lead to a reduction in capacitance and an increase in ESR over time [23].

For film capacitors, the common failure criterion is a 5% decrease in capacitance. Since the losses and temperature rise of capacitors vary significantly across different applications, manufacturer datasheets typically provide only a single reference lifetime at a specified reference hotspot temperature. When the actual hotspot temperature exceeds this reference value, the additional thermal stress will accelerate capacitor degradation. The following equation presents the capacitor lifespan model provided by the IEC standard and its commonly used simplified version [7]

$$L = L_0\left(\frac{V}{V_0}\right)^{-8} \times \exp\left[\frac{E_a}{k_0}\left(\frac{1}{T_{\text{ref}}} - \frac{1}{T_h}\right)\right] \approx L_0 \times 2^{\frac{T_{\text{ref}} - T_h}{10}} \tag{5}$$

where $L$ and $L_0$ represent the capacitor lifetime under actual and test conditions ($h$), respectively. $V$ and $V_0$ denote the operating and test voltage ($V$), while $T_{\text{ref}}$ refers to the reference temperature (°C). $E_a$ is the activation energy (kJ/mol), $k$ is the Boltzmann constant. Considering that the dielectric strength of commercial polymer films generally exceeds 250 V/$\mu$m, only capacitors that meet the voltage requirements are selected. The influence of voltage on capacitor aging can be neglected, making the temperature the primary factor affecting capacitor lifetime [15], [16]. As can be seen from (5), there is a negative exponential relationship between the lifetime and the hotspot temperature of capacitors: with the increase in the hotspot temperature, the lifetime of capacitors drops significantly [26]. For instance, the lifetime of a KEMET C4AQ capacitor is 120 000 h when the hotspot temperature is 70 °C, and 60 000 h at 85 °C. However, when operating at hotspot temperatures of 115 °C and 125 °C, its expected lifetime drops drastically to only 500 and 200 h, respectively [27]. In product manuals, these losses are represented as the power loss due to the current flowing through the $R_{\text{esr}}$. Since the ESR of capacitors increases with degradation processes, the power losses can be expressed as a function of the capacitor's damage

$$P_t = \left(\frac{I}{n}\right)^2 \times R_{\text{esr}}\left(1 + 0.23D^{2.5}\right) \tag{6}$$

where $P_t$ is the power loss of the capacitor at time $t$ ($W$), $I$ is the root mean square (rms) value of the ripple current ($A$), and $n$ is the number of capacitors connected in parallel. $D$ is damage value [3], [28]. Considering that the hotspot temperature of the capacitor varies with the acceleration and deceleration of the motor, conducting a damage accumulation analysis requires a statistical analysis of the temperature-time curve. Miner's rule is commonly used to predict component failure under alternating load conditions. It assumes that damage caused by varying stress levels can accumulate, with the order of stress application not affecting the damage process. Based on this assumption, the total cumulative damage of the capacitor under a sequence of varying stresses can be calculated through linear superposition. Thus, the damage to the capacitor can be assessed by converting the time spent at each temperature into equivalent operation time at a reference temperature and summing the results

$$D = \sum_{j=1}^{m} D_j = \sum_{j=1}^{m}\sum_{i=1}^{n}\frac{t_i}{L_i} = \sum_{j=1}^{m}\sum_{i=1}^{n}\frac{t_i}{L_0 \times 2^{\frac{T_{\text{ref}} - T_i}{10}}} \tag{7}$$

where $t_i$ represents the operating time of the capacitor at hotspot temperature $i$ in a drive cycle ($h$), $L_i$ is the expected lifetime of the capacitor at temperature $i$ ($h$) calculated using (5), and $D_j$ is the accumulated damage value in the $j$th drive cycle. After completing $m$ drive cycles, if the cumulative damage $D$ reaches 1, it indicates that the capacitor has accumulated its rated lifetime under reference conditions, meaning it has reached its failure point. The capacitor's lifetime can then be calculated as

$$L = \sum_{j=1}^{m}\sum_{i=1}^{n} t_i. \tag{8}$$

## III. ELECTRICAL AND THERMAL MODELING OF FILM CAPACITORS FOR DC-LINK APPLICATION

Given the strong correlation between capacitor aging and thermal stress, it is essential to analyze the temperature rise profile of dc-Link capacitors. In this section, Simulink, FEA, and LTN are, respectively, used to obtain the current profile, thermal impedance, and temperature rise of the capacitors, laying the foundation for subsequent lifetime evaluation.

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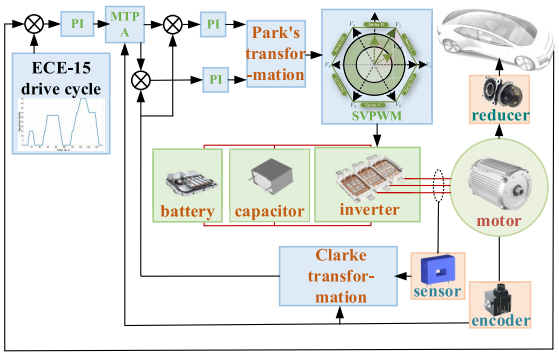



Fig. 5. Block diagram of the electrical driving system for EVs.

TABLE III
PARAMETERS USED IN SIMULINK SIMULATION

| Parameters | Values |
|---|---|
| Nominal battery voltage (V) | 317.5 |
| Capacitance (μF) | 440 (4×110) |
| Switching frequency (kHz) | 10 |
| Windward area ($A_{front}$, m$^2$) | 2.5 |
| Rolling resistance coefficient ($\mu$) | 0.019 |
| Air drag coefficient ($C_{drag}$) | 0.35 |
| Gear ratio | 9 |
| Radius of wheel ($r_{wheel}$, m) | 0.35 |
| Density of air ($\rho_a$, kg/m$^3$) | 1.225 |
| Weight of vehicle ($m$, kg) | 1800 |

### A. Electrical Modeling of Film Capacitors for EVs

Since ripple current varies with motor output torque, analyzing the EV speed profile and external resistance conditions is necessary to determine the motor's torque and the ripple current profile, which is a prerequisite for accurately obtaining the capacitor's temperature profile [21], [22]. Fig. 5 shows the block diagram of the EV drive system, which employs SVPWM to control the inverter. PI represents Proportional-Integral control, and MTPA refers to the maximum torque per ampere control strategy.

The ECE-15 driving cycle, adopted as a standardized urban test profile in Europe for evaluating household vehicle range, comprises sequential phases of uniform acceleration, deceleration, constant-speed operation, and idling phases. As illustrated in Fig. 6(a), this cycle reaches a maximum speed of 50 km/h with an average speed of approximately 19 km/h. Each 195-s cycle corresponds to a travel distance of 0.9941 km. During vehicle operation, the forces acting on the vehicle ($F$, kg·m/s$^2$) include inertial force, aerodynamic drag, and rolling resistance. Based on Table III, the motor's electromagnetic torque ($T_m$, Kg·m$^2$/s$^2$) can be quickly calculated using the following equation:

$$T_{\mathrm{m}} = \frac{T_{\mathrm{wheel}}}{\text{gear ratio}} = \frac{F \times r_{\mathrm{wheel}}}{\text{gear ratio}} = \frac{\left(\mu mg + 0.5\rho_a v^2 A_{\mathrm{front}} C_{\mathrm{drag}} + m\frac{dv}{dt}\right) \times r_{\mathrm{wheel}}}{\text{gear ratio}} \quad (9)$$

where $g$ is the acceleration due to gravity (m/s$^2$), $v$ is the velocity (m/s), and $T_{\mathrm{wheel}}$ is the torque applied to the wheel axle (kg·m$^2$/s$^2$).

The simulations were carried out in MATLAB using the ode23tb solver, with the relative tolerance set to 1e-3.

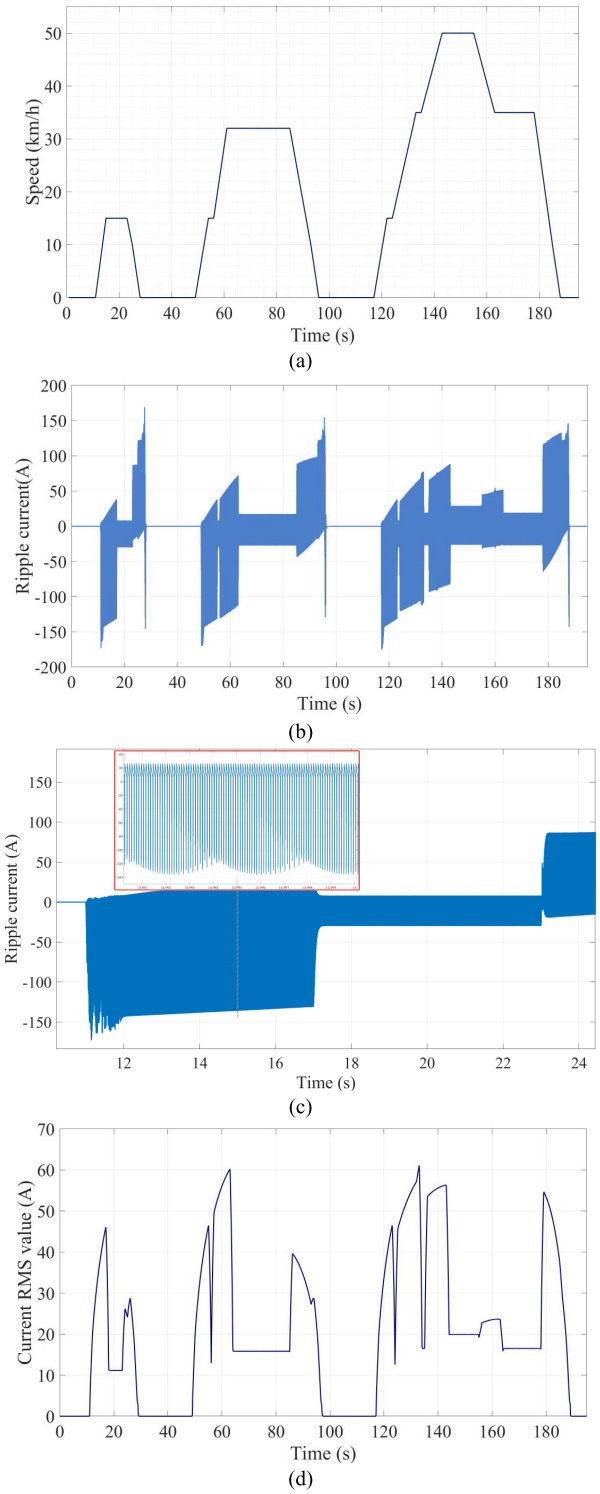



Fig. 6. Ripple current characteristics of the dc-Link capacitor. (a) ECE-15 urban driving cycle (km/h), (b) dc-link ripple current (a), (c) zoomed-in view of (b), and (d) rms value of ripple current (a).

Fig. 6(b) illustrates the simulation results of the ripple current over time, along with an enlarged view of the ripple current in Fig. 6(c). It can be observed that during the uniform acceleration phase starting at 11 s, as vehicle speed increases, aerodynamic drag gradually rises, requiring the motor output to increase accordingly to maintain constant acceleration. Because of the instantaneous increase in traction

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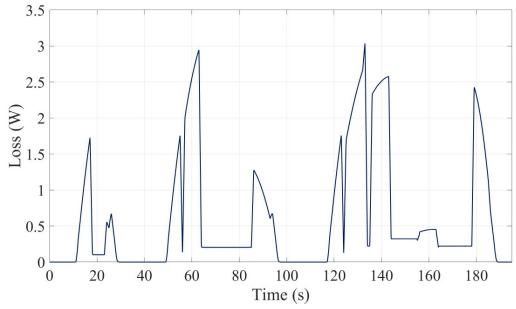



Fig. 7. Loss profile of the capacitor.

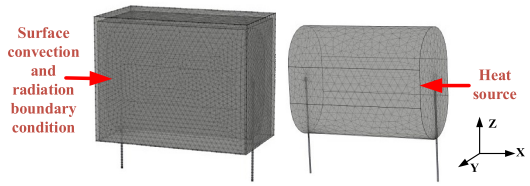



Fig. 8. Boundary conditions and mesh division of FEA simulation for the film capacitor.

power demand and the response delay of the battery current, the dc-link capacitor partially supplies the transient power by discharging. This causes the capacitor current to change from 0 A (when the vehicle is stationary) to a pulsating waveform with a dc offset, with the mean value being negative. After 15 s, when the vehicle enters the constant-speed phase, the motor output power primarily compensates for the sum of rolling resistance and aerodynamic drag and gradually converges with the battery's absorbable power. In this phase, the capacitor mainly filters the high-frequency current ripple generated by inverter switching, thereby stabilizing the dc bus voltage, so its current gradually decreases to a small range. During the deceleration phase beginning at 23 s, the motor operates in braking mode, converting mechanical energy into electrical energy that is fed back to the dc bus via the inverters. Because of limitations in battery current dynamics, the battery cannot immediately absorb all the regenerated energy. The capacitors, therefore, absorb the returned energy, resulting in a positively biased pulsating current.

Figs. 6(d) and 7 further show the rms current and the corresponding capacitor loss that iscalculated by multiplying the rms current by the ESR according to (1). It can be observed that capacitor losses are higher during the acceleration and deceleration phases, while remaining relatively constant during the constant-speed phase. Moreover, the magnitude of the loss increases with vehicle speed. Since the capacitor bank in the simulation consists of four capacitors connected in parallel, the peak rms current through a single capacitor is about 15 A.

### *B. Finite Element Modeling of Film Capacitors*

The FEA model is a numerical method that simplifies complex continuum problems by discretizing the domain into triangular or quadrilateral elements. It assumes that physical quantities vary linearly or quadratically within each element. By solving the assembled equations, the distribution of the physical field throughout the domain can be obtained. Based on the parameters in Table I, an FEA model of the C3D capacitor shown in Fig. 2 was developed in COMSOL Multiphysics. Fig. 8 shows the model's boundary conditions and mesh distribution. The capacitor core is defined as the heat source, while the external surfaces of the capacitor are assigned convective and radiative boundary conditions. The convective heat transfer coefficient and surface emissivity are set to 9.49 W/(m$^2$·°C) and 0.85, respectively [29]. In the two simulations, the ambient temperature is set to 70 °C and 60 °C, respectively. The model consists of 57 566 triangular mesh elements. The heat flux can then be calculated as

$$q = \nabla_T \cdot \varepsilon\sigma\left((T_a + 273.15)^4 - (T_c + 273.15)^4\right) + h(T_a - T_c) \tag{10}$$

where $q$ is the heat flux per unit area [J/(m$^2$·s)], $\varepsilon$ is the surface emissivity, a physical parameter that quantifies the ability of a material's surface to emit energy in the form of thermal radiation, $\sigma$ is the Stefan–Boltzmann constant, $T_a$ and $T_c$ are the case and ambient temperatures (°C), respectively, $h$ is the convective heat transfer coefficient [W/(m$^2$·°C)] reflects the intensity of convective heat exchange, defined as the heat transferred per unit area per unit time for a 1 °C temperature difference between the surface and surrounding fluid.

The material parameters are listed in Table IV [8]. Since the thickness of a single polymer film layer varies depending on the rated voltage of the capacitors, there is no standard thermal conductivity value for the capacitor core. A widely adopted approach is to model the capacitor core using an analogy with series–parallel combinations of thermal resistances [2], [8], [30]. As shown in Fig. 9, when heat flows along the radial direction, the thermal resistances of the polymer films and metallized electrodes are connected in series, whereas in the axial direction, they are arranged in parallel. Accordingly, the thermal conductivities of the capacitor core, denoted as [$k_f$, W/(m·K)] in the two directions can be expressed as follows:

$$k_f = \begin{Bmatrix} \dfrac{k_p \cdot k_m \cdot (H_p + H_m)}{k_p \cdot H_m + k_m \cdot H_P} & \text{(radial)} \\ \dfrac{k_p \cdot H_p + k_m \cdot H_m}{H_p + H_m} & \text{(axial)} \end{Bmatrix} \tag{11}$$

TABLE IV
MATERIALS' PARAMETERS OF THE CAPACITOR

| Parameters | Values |
|---|---|
| Thickness (mm) | $5\times10^{-3}$ (film); $20\times10^{-6}$ (electrode) |
| Thermal conductivity (W/(m·K)) | 0.22 (film); 237 (electrode); 1.1633 (core-x); 0.2209 (core-y/z); 0.15 (resin); 15 (PPS); 400(copper) |
| Density (kg/m$^3$) | $0.9\times10^3$ (film); $2.7\times10^3$ (electrode); 907.17 (core); 1820 (resin); 1650 (PPS); 8960(copper) |
| Specific heat capacity (J/(kg·°C)) | $1.9\times10^3$ (film); $0.9\times10^3$ (electrode); 1888.14 (core); 1340 (resin); 1000 (PPS); 385(copper) |

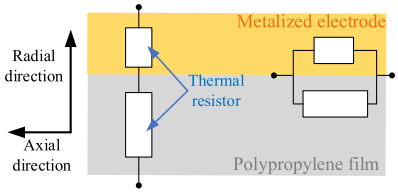
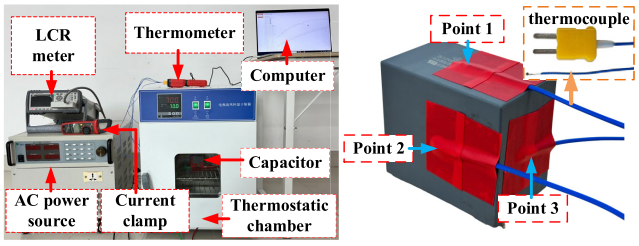



Fig. 9. Anisotropy of thermal resistance in capacitor core.

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TABLE V
TEMPERATURE RISE OF THE CAPACITOR

| Temperature rise | Simulation (in °C) | Experiment (in °C) |
|---|---|---|
| Hotspot | 7.0 ($T_a$=70°C) | 7.1 |
| | 7.1 ($T_a$=60°C) | (Calculated by (2)) |
| Casing Point 1 | 2.6 ($T_a$=70°C) | 2.4 ($T_a$=70°C) |
| | 2.7 ($T_a$=60°C) | 2.5 ($T_a$=60°C) |
| Casing Point 2 | 2.8 ($T_a$=70°C) | 2.9 ($T_a$=70°C) |
| | 2.9 ($T_a$=60°C) | 3.0 ($T_a$=60°C) |
| Casing Point 3 | 2.9 ($T_a$=70°C) | 2.9 ($T_a$=70°C) |
| | 3.0 ($T_a$=60°C) | 2.9 ($T_a$=60°C) |




(a) (b)

Fig. 10. Experimental setup for evaluating the temperature rise of film capacitors. (a) Experimental equipment. (b) Thermocouples used and their installation positions.

where $k_p$ and $k_m$ represent the thermal conductivities of the polymer film and the electrode, respectively, while $H_p$ and $H_m$ represent the thickness of the polymer film and the electrode ($m$) in Table V, respectively. For the selected capacitor, the thickness of the polymer film is 250 times that of the metallized electrode, while its thermal conductivity is much lower than that of the electrode. Considering the configuration shown in Fig. 9, the thermal conductivity of the capacitor core in the radial direction is primarily determined by the polymer, whereas in the axial direction, it is significantly influenced by the metal electrode. For the capacitor core model in Fig. 8, the thermal resistances of the polymer film and metal electrode are in series along the $y$-axis and $z$-axis, while along the $x$-axis, the two resistances are in parallel. Therefore, by using COMSOL's capability to define anisotropic material properties, the results from (11) are applied independently along the three directions. The density ($\rho_f$, kg/m$^3$) and specific heat capacity [$C_f$, J/(kg·°C)] of the capacitor core are calculated using the following equations, respectively [2], [30]:

$$\rho_f = \frac{\rho_p \cdot H_p + \rho_m \cdot H_m}{H_p + H_m} \tag{12}$$

$$C_f = \frac{C_p \cdot \rho_p \cdot H_p + C_m \cdot \rho_m \cdot H_m}{\rho_p \cdot H_p + \rho_m \cdot H_m} \tag{13}$$

where $\rho_p$ and $\rho_m$ are the density of the metallized film, polymer film, and metal electrode (kg/m$^3$), $C_p$ and $C_m$ are the specific heat capacity of polymer film and metal [J/(kg·°C)].

To validate the accuracy of the FEA, capacitor temperature rise experiments were conducted in the laboratory. Fig. 10(a) shows the experimental platform used for measuring the capacitor's temperature rise [31]. Since the purpose of injecting current is to generate heat loss for FEA validation, any ac source within the capacitor's voltage rating can be used. A 110 $\mu$F C3D capacitor was directly connected to a 220 V, 50 Hz ac source. The ESR at this frequency (13 m$\Omega$) was measured using a Tonghui 2832 LCR meter, and the current (7.4 A) was measured with a UNI-T UT202 current clamp. Based on these values, the internal power dissipation was calculated to be 0.714 W. The capacitor was placed inside a Shangcheng 101–00BS thermostatic chamber, with ambient temperatures set to 70 °C and 60 °C for two separate tests. Because of structural constraints, the hotspot temperature inside the capacitor cannot be measured directly without damaging it. Thus, only the casing surface temperature was recorded. A UNI-T UT535F thermocouple thermometer with ±0.4% accuracy was used to measure temperatures at various locations on the capacitor surface. Fig. 10(b) shows the thermocouple sensors and their attachment points on the capacitor surface. The thermocouples were fixed to the capacitor surface using adhesive and secured with tape. Since the internal hotspot cannot be directly measured, the hotspot temperature obtained from the simulation was compared with calculated reference values using (2), thermal resistance data from Table I, and the loss value.

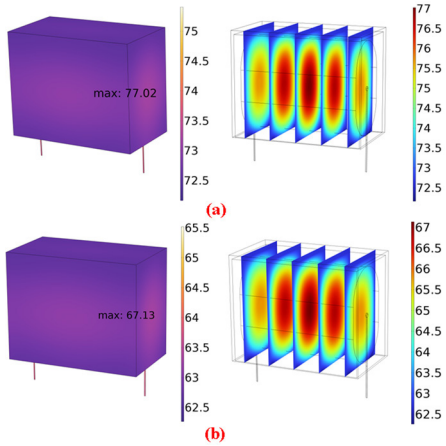



Fig. 11. Temperature distribution of the capacitor casing and internal core obtained by FEA (°C). (a) Simulation result at ambient temperature of 70 °C. (b) Simulation result at ambient temperature of 60 °C.

Fig. 11 presents the FEA simulation results, with internal power loss set to 0.714 W. Given that the core volume is 65 903 mm$^3$, the corresponding power loss density is $1.08 \times 10^{-5}$ (W/mm$^3$). The Virtual Probes tool is used to monitor the temperature rise at the hottest point inside the capacitor. The maximum temperatures reached 77.0 °C and 67.1 °C, respectively, in the two tests. As shown in Fig. 11, the temperature gradient along the direction of point 3 in Fig. 10(b) is larger than those at points 1 and 2. This is due to the internal arrangement of polymer films and electrodes illustrated in Fig. 9, which results in different thermal conductivities in various directions.

The simulation, theoretical calculation, and experimental results are summarized in Table V [22], with all results rounded to one decimal place. The simulated temperature rise on the capacitor casing surface ranges from 2.4 °C to 3.0 °C, which corresponds well with the measured range of 2.6 °C–3.0 °C. The maximum errors for hotspot and casing temperatures are 1.71% and 7.69%, respectively. For hotspot temperature, the value calculated using the reference thermal resistance and power loss is 7.1 °C, while FEA results

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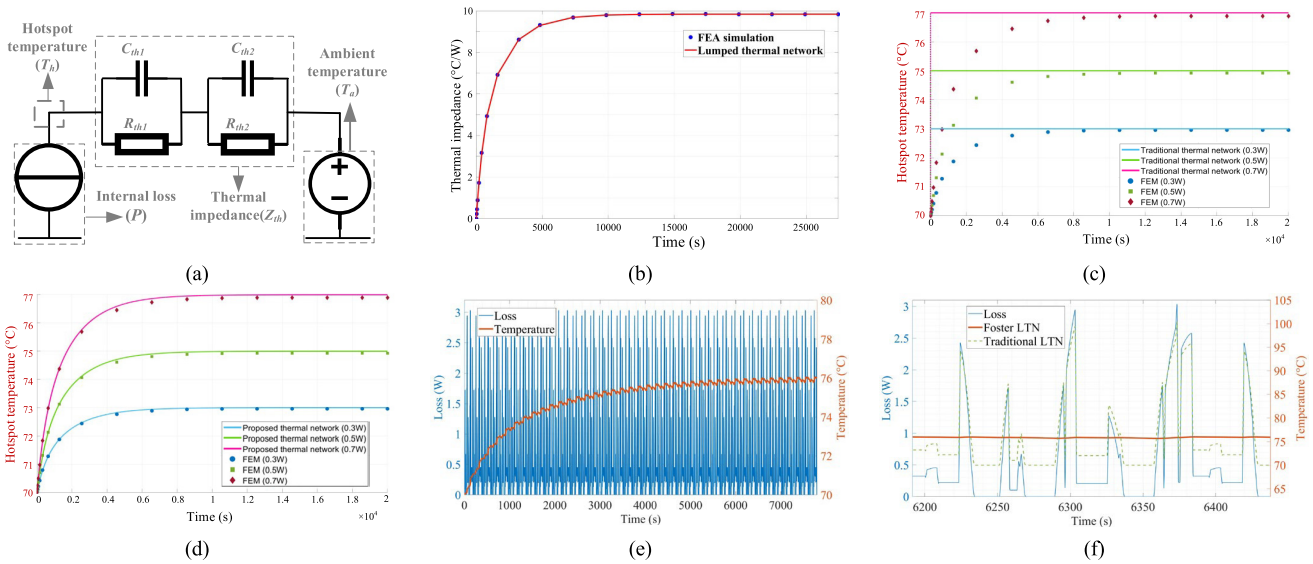



Fig. 12. Proposed LTN of the capacitor and simulation results. (a) LTN topology. (b) Thermal impedance characteristics. (c) Temperature rise of capacitors with different loss values obtained with traditional LTN. (d) Temperature rise of capacitors with different loss values obtained with the proposed LTN. (e) Change of hotspot temperature of the capacitor over 40 ECE-15 cycles. (f) Temperature rise profiles obtained with different LTNs.

are 7.1 °C and 7.0 °C under ambient conditions of 60 °C and 70 °C, confirming good agreement with theory.

### C. LTN of Film Capacitors

Since conducting FEA over long time scales demands substantial computational resources, such models are not suitable for assessing the temperature rise of capacitors under time-varying operating conditions. Therefore, it is essential to translate FEA results into an LTN, which serves as a numerical model capable of rapidly calculating the capacitor's hotspot temperature based on internal power losses and ambient temperature over extended durations.

The traditional LTN of capacitors shown in Fig. 3(b) completely neglects the effect of thermal capacitance and thus significantly deviates from actual temperature rise behavior. To overcome this limitation, the Foster thermal network, commonly used in power module studies and illustrated in Fig. 12(a), is adopted to characterize the thermal behavior of the capacitor [28]. The thermal impedance is composed of a series of thermal resistances ($R_{\text{thi}}$) and thermal capacitances ($C_{\text{thi}}$). Thermal resistance describes a material's opposition to heat flow, while thermal capacitance reflects its ability to store thermal energy, defined as the heat required to raise the temperature by 1 °C. The parameters of the thermal network can be calculated as follows:

$$Z_{\text{th}}(t) = \frac{T_h(t) - T_a}{P} = \sum_{i=1}^{2} R_{\text{th}_i}\left(1 - e^{\frac{1}{C_{\text{th}_i} \times R_{\text{th}_i}}}\right) \tag{14}$$

where $Z_{\text{th}}$ is the thermal impedance between the hotspot and the environment (°C/W), $R_{\text{thi}}$ (°C/W), and $C_{\text{thi}}$ (J/°C) are the thermal resistance and capacitance of order $i$, respectively.

As can be seen from Fig. 12(b), the network parameters are extracted by fitting transient thermal impedance values from FEA simulations using (14): $C_{\text{th1}}$ and $C_{\text{th2}}$ are 142.8745 and 327.2810 (J/°C), respectively. $R_{\text{th1}}$ and $R_{\text{th2}}$ are 3.7506 and 6.0874 (°C/W), respectively. The sum of the second-order thermal resistances in the thermal network is 9.838, showing only a 1.62% deviation from the manufacturer-provided value of 10 in Table I [24].

Based on the proposed thermal network, the initial temperature rise ($T_{h1}$,°C) with an internal power loss value ($P_1$, W) of the capacitor can be expressed as (15). This approach is analogous to electrical circuit theory, where the total response (hotspot temperature) is the sum of a steady-state component (ambient temperature) and a transient component (self-heating) that decays over time

$$T_{h1} = P_1 \times \sum_{i=1}^{2} R_{\text{th}_i}\left(1 - e^{\frac{1}{C_{\text{th}_i} \times R_{\text{th}_i}}}\right) + T_{\text{a}}. \tag{15}$$

When the internal power loss of the capacitor changes to ($P_2$, W) at the next second, based on the superposition theorem of circuits, this is equivalent to superimposing a new transient response onto the original total response: the hotspot temperature ($T_{h2}$,°C) can be expressed as follows:

$$T_{h2} = P_2 \times \sum_{i=1}^{2} R_{\text{th}_i}\left(1 - e^{\frac{1}{C_{\text{th}_i} \times R_{\text{th}_i}}}\right) + (T_{h1} - T_{\text{a}}) \sum_{i=1}^{2} R_{\text{th}_i} e^{\frac{1}{C_{\text{th}_i} \times R_{\text{th}_i}}} + T_{\text{a}}. \tag{16}$$

On this basis, the hotspot temperature rise of the capacitor at the $k$th second ($T_{hk}$,°C) is as follows:

$$T_{hk} = \sum_{j=1}^{k}\left[P_j \times \sum_{i=1}^{2} R_{\text{th}_i}\left(e^{\frac{-(k-j)}{C_{\text{th}_i} \times R_{\text{th}_i}}} - e^{\frac{-(k+1-j)}{C_{\text{th}_i} \times R_{\text{th}_i}}}\right)\right] + T_{\text{a}}. \tag{17}$$

Fig. 12(c) and (d) compares the heating behavior of the capacitor predicted by FEA, the traditional LTN, and the proposed LTN under losses of 0.3, 0.5, and 0.7 W. The temperature rise trends obtained by FEA and the proposed LTN show a high degree of consistency, with the transient temperature rises at various points in the FEA simulation closely matching the curves calculated using (17). In contrast, the traditional

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LTN model shows unrealistic behavior, with an instantaneous jump to the steady-state temperature when $t = 0$. This comparison further validates the accuracy and importance of the proposed LTN. Regarding efficiency, on the given computer setup (12th Gen Intel i7-1260P processor at 2.10 GHz with 16.0 GB RAM), COMSOL's FEA takes 45–46 s to solve the temperature rise for a single capacitor. In contrast, MATLAB displays that both traditional and proposed LTNs complete the calculation within 0.01 s, highlighting the proposed method's high computational efficiency. Fig. 12(e) presents the hotspot temperature profile of the capacitor over 40 ECE-15 driving cycles based on (17). Fig. 12(f) further illustrates the data within the range of 6200–6500 s, with additional results obtained from the conventional thermal network based on (2) for comparison. The proposed method yields results that better correspond to physical expectations: in the initial stage, the hotspot temperature gradually increases during periods of high ripple current and decreases when the ripple current is zero. Moreover, as the temperature difference between the capacitor hotspot and the ambient environment increases, the temperature rise caused by the same power loss diminishes, the rate of cooling becomes larger, and the capacitor temperature decreases under lower or zero loss conditions, eventually stabilizing within a narrow fluctuation range. By contrast, the traditional method produces instantaneous temperature oscillations between 70 °C and 101 °C, with several abrupt variations exceeding 10 °C during acceleration and braking events.

Although the proposed method demonstrates good accuracy in Fig. 12(d) and (e), potential sources of error remain and may stem from the following factors.

1) To improve computational efficiency, the surface emissivity and convective heat transfer coefficient in the FEA were set to typical values commonly used in capacitor thermal studies; however, these parameters may vary under different cooling conditions.
2) Thermal properties such as conductivity were assumed constant in the simulation, while materials like metal pins often exhibit temperature-dependent behavior, which can impact accuracy.
3) A second-order LTN was employed; although it effectively captures the transient thermal response shown in Fig. 10, higher-order models may be required to accurately reflect temperature variations under dynamically changing loss conditions.

## IV. Reliability Assessment of Film Capacitors Considering Manufacturing Tolerance

Based on the cumulative damage model and the temperature profile obtained in the previous section, the lifetime of the capacitor is assessed. A Monte Carlo simulation is conducted to evaluate the impact of manufacturing tolerances on the capacitor, enabling the determination of B10 life to support maintenance scheduling.

### A. Lifetime Evaluation of Film Capacitors

Fig. 13 illustrates the steps of evaluating the capacitor lifetime, which considers both task profiles and parameter aging. This method consists of four parts: parameter input, electrical-thermal modeling, damage accumulation, and reliability evaluation.

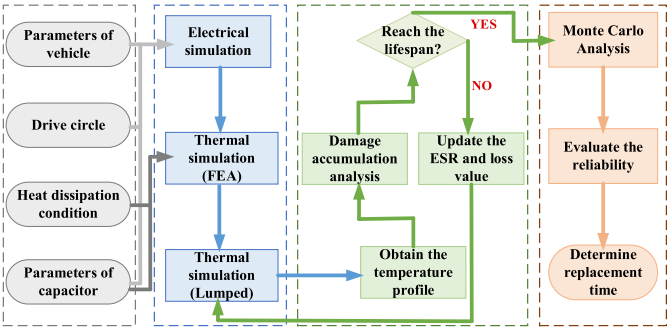



Fig. 13. Process of lifetime evaluation for metallized film capacitors used in EVs.

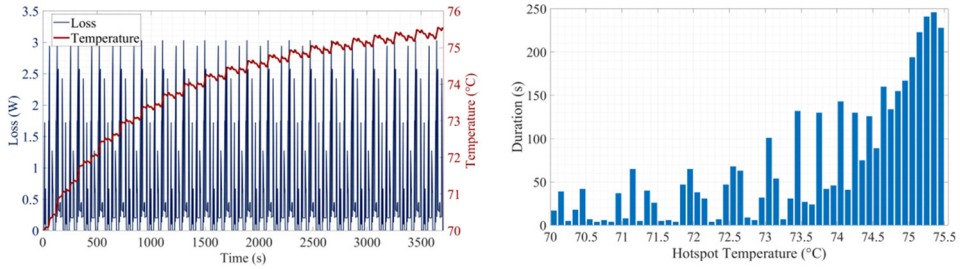
First, the driving cycle and the parameters of the vehicle and capacitors are imported into Simulink to obtain the current profile. Meanwhile, a FEA model is developed based on the capacitor specifications and cooling conditions to extract the parameters of the thermal network. Then, the power loss and hotspot temperature profiles are calculated using (6) and (16). By tracking the hotspot temperature throughout the driving cycle, the operating time at each temperature level is recorded, allowing cumulative damage to be evaluated using (7). If the accumulated damage is less than 1, it is substituted into (6) to update the power losses. The total operating time is then calculated using (8). Considering the computational cost of updating losses in every cycle and the slow rate of capacitor aging, losses are updated only when cumulative damage increases by 0.1. These updated values are used for temperature and damage calculations in subsequent cycles. When the accumulated damage reaches 1, the capacitor lifetime is determined.

Using this method, four 110 $\mu$F C3D capacitors connected in parallel were evaluated, and the ambient temperature was set at 70 °C [6], [32]. Based on the approach proposed in Section III, the current profile and thermal network were constructed. According to the National Household Travel Survey, the average car trip length in 2022 was 18.49 km [33]. Thus, 19 driving cycles were set to represent a single trip. The resulting temperature rise profile and the distribution of operating time at different hotspot temperatures are shown in Fig. 14(a) and (b). The estimated capacitor lifetime was 74475.1 h, representing a reduction of about 35.5% compared with the 100 000-h reference lifetime at the reference hotspot temperature (70 °C) listed in Table I. This result indicates that the additional temperature rise caused by the capacitor's internal losses can significantly accelerate its aging, especially in applications with relatively high ambient temperatures such as traction inverters.

### B. Reliability Evaluation of Film Capacitors

Obviously, the predicted lifespan above represents a fixed value, assuming all capacitors fail at the same time. However, due to manufacturing tolerances, even capacitors with the same rated specifications can exhibit varying lifetimes. Given that capacitor failures can pose greater risks than replacement costs, it is advisable to replace capacitors once reliability drops below a certain threshold to maintain system safety [33]. Therefore, fixed lifetime predictions must be converted into statistical values that account for parameter variations.

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(a) (b)

Fig. 14. Capacitor hotspot temperature variation and duration distribution in a single trip. (a) Variation in hotspot temperature during operation (°C). (b) Cumulative operation time under different temperatures (s).

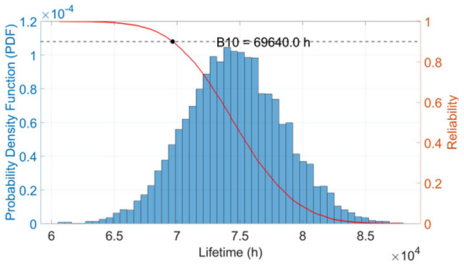



Fig. 15. Distribution of the lifetime of film capacitors and the change in their reliability over time.

To evaluate the influence of manufacturing tolerances, the Monte Carlo method is applied. Because of the dynamic variation in hotspot temperature, it is difficult to introduce parameter deviations directly into the transient simulation. Instead, an equivalent static temperature ($T_e$,°C) is calculated based on the following equation and lifetime:

$$\frac{T_{\text{ref}} - T_e}{10} = \log_2\left(\frac{L}{L_0}\right). \tag{18}$$

In this case, the equivalent hotspot temperature is 74.25 °C. Based on the 10% capacitance variation range of adopted capacitor products, 10 000 samples are generated in MATLAB, with key parameters including rated lifetime and hotspot temperature rise following a Gaussian distribution with a 10% standard deviation. The failure distribution is then fit using a Weibull function, and its cumulative density function is used to assess capacitor reliability [16]

$$R(t) = e^{-\left(\frac{t}{\eta}\right)^{\beta}} \tag{19}$$

where $\beta$ is the shape parameter indicating the failure mode, $\eta$ is the scale parameter, and $t$ is the operating time ($h$).

Considering the significant risks associated with capacitor failure, researchers commonly use the B10 lifetime (i.e., the time by which 10% of the capacitors have failed, corresponding to 90% reliability) to guide maintenance decisions. As shown in Fig. 15, most failures occur between 65 000 and 85 000 h, with the calculated B10 life being 69 640 h. Compared to the static lifetime of 74475.1 h (corresponding to the time when 63.2% of capacitor samples fail), 53.2% of samples fail within the next 4835.1 h after reaching the B10 threshold. This highlights the need to base maintenance policies on B10 life rather than static lifetime. Furthermore, considering that the consequences of capacitor failure are typically more severe than the cost of early replacement, and potential discrepancies between empirical model parameters and real-world operating conditions across capacitor types, it is advisable to adopt a conservative replacement policy. For the capacitor evaluated under these conditions, a conservative replacement interval between 65 000 and 70 500 h is recommended.

Future work can be conducted on two main aspects: First, both the FEA and the LTN models can be extended to account for thermal coupling among capacitors and other inverter components to improve the accuracy of lifetime prediction and maintenance scheduling. Second, the impact of different control and modulation strategies on ripple current and temperature can be studied to identify optimized approaches that reduce thermal stress and extend capacitor lifetime.

## V. Conclusion

This article proposes a temperature rise assessment method for dc-link film capacitors in traction inverters of EVs. Based on the ripple current profile obtained through a Simulink model and the thermal impedance derived from FEA, a second-order LTN is used to determine the temperature rise profile of the capacitor. Compared to traditional pure thermal resistor LTNs, this LTN effectively addresses the issue of instantaneous temperature fluctuations caused by neglecting the capacitor's heat capacity. On this basis, a reliability evaluation is conducted to consider the impact of manufacturing tolerances on capacitor lifetime. The B10 life obtained can serve as a reference for decision-making regarding the replacement and maintenance of capacitors used in traction inverters of EVs.

## References

[1] H. Wen, W. Xiao, X. Wen, and P. Armstrong, "Analysis and evaluation of DC-link capacitors for high-power-density electric vehicle drive systems," *IEEE Trans. Veh. Technol.*, vol. 61, no. 7, pp. 2950–2964, Sep. 2012.

[2] Z. Wang, F. Yan, Z. Hua, L. Qi, Z. Hou, and Z. Xu, "Geometric optimization of self-healing power capacitor with consideration of multiple factors," *J. Power Sources*, vol. 323, pp. 147–157, Aug. 2016.

[3] K. Kuang, X. Guo, C. Li, and X. Li, "A novel lifetime estimation method and structural optimization design for film capacitors in EVs considering material aging and power losses," *IEEE Trans. Device Mater. Rel.*, vol. 24, no. 3, pp. 365–379, Sep. 2024.

[4] B. Zhang, Z. Zhao, Y. Liu, H. Ma, C. Shi, and Y.-H. Mei, "Reliable WBG devices packaging by forming nano-gradient structures on silver-plated substrates," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 14, no. 7, pp. 1156–1163, Jul. 2024.

[5] H. Wang and F. Blaabjerg, "Reliability of capacitors for DC-link applications in power electronic converters—An overview," *IEEE Trans. Ind. Appl.*, vol. 50, no. 5, pp. 3569–3578, Sep. 2014.

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[6] S. Yang, A. Bryant, P. Mawby, D. Xiang, L. Ran, and P. Tavner, "An Industry-based survey of reliability in power electronic converters," *IEEE Trans. Ind. Appl.*, vol. 47, no. 3, pp. 1441–1451, May 2011.

[7] *Electric Components Reliability Reference Conditions for Failure Rates and Stress Models for Conversion*, Standard IEC 61709, 2017.

[8] K. Kuang, X. Guo, C. Li, X. Xi, and H. Fang, "A novel multiscale perspective based hotspot temperature assessment method for film capacitor in DC-link applications," *IEEE Trans. Ind. Appl.*, vol. 60, no. 9, pp. 9111–9122, Sep. 2024.

[9] K. Kuang, X. Guo, Z. Zhou, C. Li, and X. Li, "Mission profile-based hotspot temperature and lifespan estimation of DC-link capacitors used in automotive traction inverters," *IEEE Trans. Device Mater. Rel.*, vol. 25, no. 1, pp. 134–143, Mar. 2025.

[10] H. Niu, H. Li, and S. Wang, "Improved thermal simulation of aluminum electrolytic capacitor for lifetime prediction," in *Proc. IEEE 9th Int. Power Electron. Motion Control Conf. (IPEMC-ECCE Asia)*, Nanjing, China, Nov. 2020, pp. 1838–1843.

[11] Z. Yin, R. Wu, M. Hygum, Y. Yang, and H. Wang, "Thermal stress model of transversal forced air-cooled capacitor banks for MW power converters," in *Proc. 10th Int. Conf. Power Electron. ECCE Asia*, May 2019, pp. 3096–3101.

[12] M. Wang, W. Jiang, and K. Ma, "Circulating current controls for optimal thermal stress of capacitors in MMC systems," *IEEE Trans. Power Electron.*, vol. 39, no. 10, pp. 12435–12445, Oct. 2024.

[13] Z. Yin, Y. Yang, and H. Wang, "Thermal modeling of an electrolytic capacitor bank," in *Proc. IEEE 21st Workshop Control Model. Power Electron. (COMPEL)*, Aalborg, Denmark, Nov. 2020, pp. 1–5.

[14] H. Wang, R. Zhu, H. Wang, M. Liserre, and F. Blaabjerg, "A thermal modeling method considering ambient temperature dynamics," *IEEE Trans. Power Electron.*, vol. 35, no. 1, pp. 6–9, Jan. 2020.

[15] Y. Yang, K. Ma, H. Wang, and F. Blaabjerg, "Instantaneous thermal modeling of the DC-link capacitor in PhotoVoltaic systems," in *Proc. IEEE Appl. Power Electron. Conf. Expo. (APEC)*, Charlotte, NC, USA, Mar. 2015, pp. 2733–2739.

[16] D. Zhou, Y. Song, Y. Liu, and F. Blaabjerg, "Mission profile based reliability evaluation of capacitor banks in wind power converters," *IEEE Trans. Power Electron.*, vol. 34, no. 5, pp. 4665–4677, May 2019.

[17] S. Huang, H. Wang, D. Kumar, G. Zhu, and H. Wang, "Reliability evaluation of DC-link capacitors in multi-drive systems," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, Baltimore, MD, USA, Sep. 2019, pp. 3157–3162.

[18] H. Wang, P. Davari, D. Kumar, F. Zare, and F. Blaabjerg, "The impact of grid unbalances on the reliability of DC-link capacitors in a motor drive," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, Cincinnati, OH, USA, Oct. 2017, pp. 4345–4350.

[19] J. M. Lenz, J. R. Pinheiro, and H. C. Sartori, "DC-link electrolyte capacitor lifetime analysis for a PV boost converter," in *Proc. IEEE 8th Int. Symp. Power Electron. Distrib. Gener. Syst. (PEDG)*, Florianopolis, Brazil, Apr. 2017, pp. 1–6.

[20] H. Wang, P. Diaz Reigosa, and F. Blaabjerg, "A humidity-dependent lifetime derating factor for DC film capacitors," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, Montreal, QC, Canada, Sep. 2015, pp. 3064–3068.

[21] B. Yao, X. Ge, H. Wang, H. Wang, D. Zhou, and B. Gou, "Multi-timescale reliability evaluation of DC-link capacitor banks in metro traction drive system," *IEEE Trans. Transport. Electrific.*, vol. 6, no. 1, pp. 213–227, Mar. 2020.

[22] B. Yao et al., "Electrothermal stress analysis and lifetime evaluation of DC-link capacitor banks in the railway traction drive system," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 4, pp. 4269–4284, Aug. 2021.

[23] Z. Zhao, P. Davari, W. Lu, H. Wang, and F. Blaabjerg, "An overview of condition monitoring techniques for capacitors in DC-link applications," *IEEE Trans. Power Electron.*, vol. 36, no. 4, pp. 3692–3716, Apr. 2021.

[24] *Faratronic: DC-Link Capacitor for PCB_C3D*. Accessed: Oct. 25, 2025. [Online]. Available: http://files.faratronic.com/book/2022/C3D.pdf

[25] H. Tong, W. Yao, C. Li, H. Luo, and W. Li, "Current sharing analysis and evaluation of parallel DC-link capacitors in vehicle motor drive," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 12, no. 2, pp. 2188–2202, Apr. 2024.

[26] A. A. Lad, E. Roman, Y. Zhao, W. P. King, and N. Miljkovic, "Fin design topology optimization for direct liquid cooling of multichip power modules," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 14, no. 5, pp. 795–809, May 2024.

[27] KEMET. *C4AQ Application Note*. Accessed: Oct. 25, 2025. [Online]. Available: https://content.kemet.com/datasheets/KEM_F3114_C4AQ.pdf

[28] H. Wang, P. Davari, H. Wang, D. Kumar, F. Zare, and F. Blaabjerg, "Lifetime estimation of DC-link capacitors in adjustable speed drives under grid voltage unbalances," *IEEE Trans. Power Electron.*, vol. 34, no. 5, pp. 4064–4078, May 2019.

[29] T. Lledo-Ponsati, A. S. Bahman, F. Iannuzzo, D. Montesinos-Miracle, and S. Galceran-Arellano, "Thermal modeling of large electrolytic capacitors using FEM and considering the internal geometry," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 5, pp. 6315–6328, Oct. 2021.

[30] Z. Wang et al., "Optimal design of high temperature metalized thin-film polymer capacitors: A combined numerical and experimental method," *J. Power Sources*, vol. 357, pp. 149–157, Jul. 2017.

[31] Z. Wang, Y. Lu, and Y. Ma, "Temperature rise of metallized film capacitors under AC and DC superimposed voltage," *IEEE Trans. Plasma Sci.*, vol. 49, no. 12, pp. 3883–3891, Dec. 2021.

[32] Y. Lu et al., "A 3-D temperature-dependent thermal model of IGBT modules for electric vehicle application considering various boundary conditions," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 12, no. 6, pp. 5463–5475, Dec. 2024.

[33] *National Household Travel Survey*. Accessed: Oct. 25, 2025. [Online]. Available: https://nhts.ornl.gov/



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